

Quantum-Inspired Annealing for Multi-Stage Reasoning in Small Language Models: A Literature Review

Student Team

Samsung PRISM Programme — Worklet 26ARS05

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Abstract— Large Language Models (LLMs) have achieved impressive performance on a wide range of natural-language tasks, but their ability to perform robust multi-step reasoning remains limited, particularly in the Small Language Model (SLM) regime. This paper reviews the recent literature on reasoning in LLMs and on combinatorial-optimisation-based reasoning selection. We organise the survey into three parts: (i) the broad LLM-reasoning landscape and the motivation for structured prompting; (ii) recent variants of Chain-of-Thought (CoT) prompting, including Self-Consistency, Tree-of-Thoughts, ReAct, Ranked-Voting Self-Consistency, Combinatorial Reasoning and Quantum Combinatorial Reasoning; and (iii) Quadratic Unconstrained Binary Optimisation (QUBO) as a mathematical substrate for reasoning-fragment selection. We close by mapping each stage of the proposed worklet pipeline to its closest published antecedent, motivating an SLM-plus-quantum-inspired-annealer architecture for the Samsung PRISM 26ARS05 worklet.

Index Terms— Large language models, small language models, chain-of-thought, reasoning, quadratic unconstrained binary optimisation, quantum annealing, simulated annealing, combinatorial reasoning.

I. INTRODUCTION

Recent Large Language Models (LLMs) such as GPT-3, GPT-4 and the Llama family have demonstrated state-of-the-art performance across a broad range of natural-language processing (NLP) tasks [1]. Despite this progress, tasks that require explicit multi-step reasoning—arithmetic word problems, symbolic manipulation, multi-hop question answering, and planning—continue to expose weaknesses even in the largest available models [2], [6].

The comprehensive survey of Liu et al. [1] traces the full lifecycle of modern LLMs from training to inference and observes that purely increasing model size yields diminishing returns on reasoning benchmarks. Consequently, the research community has shifted attention to (i) structured prompting strategies that elicit intermediate reasoning, (ii) aggregation strategies that combine multiple

sampled reasoning traces, and (iii) optimisation-based methods that select among candidate reasoning fragments rather than relying on a single decoded chain [5], [8], [9].

This paper reviews the literature relevant to the Samsung PRISM worklet 26ARS05, which targets a Quantum-Inspired Annealing pipeline for Multi-Stage Reasoning over Small Language Models (SLMs). The remainder is organised as follows. Section II reviews LLM reasoning and the rationale for working with SLMs. Section III surveys Chain-of-Thought (CoT) prompting algorithms and their recent variants. Section IV introduces Quadratic Unconstrained Binary Optimisation (QUBO) and its role in modern reasoning-selection pipelines. Section V maps the proposed pipeline to prior work and identifies the novel contributions of the worklet. Section VI concludes.

II. LARGE LANGUAGE MODELS AND REASONING

A. Why Reasoning Is Hard for LLMs

Autoregressive LLMs are trained to predict the next token given the previous context; they have no explicit mechanism for back-tracking, sub-goal management or verification of intermediate results. As a consequence, when a single decoding error occurs early in a long reasoning chain it tends to propagate to the final answer. Kojima et al. [3] showed that simply prepending the phrase “Let’s think step by step” to a prompt allows even a sufficiently large LLM to act as a competent zero-shot reasoner, which strongly suggests that the latent reasoning ability already exists in the model and merely needs to be elicited through prompting.

B. Beyond Scale: Structured Reasoning

Liu et al. [1] argue that scaling alone is insufficient for reliable reasoning, and that the field is converging on three complementary techniques: structured prompting (Section III), reasoning aggregation (Self-Consistency [5] and its descendants), and external action loops (ReAct [4]). Each of these techniques can be viewed as imposing structure on the otherwise unconstrained generation process.

C. Small Language Models

Small Language Models (SLMs) in the 1–3 billion parameter regime, such as Llama-3.2-3B-Instruct and TinyLlama-1.1B-Chat, are attractive for on-device

deployment, but they reason less reliably than their 70 B+ counterparts. The central hypothesis of the proposed worklet is that combining an SLM with a principled reasoning-selection mechanism can recover much of the accuracy gap while retaining the favourable footprint of an SLM. The remainder of this paper reviews the algorithmic ingredients required to realise this hypothesis.

III. CHAIN-OF-THOUGHT VARIANTS AND PROMPTING ALGORITHMS

We survey nine prompting and reasoning algorithms that have appeared in the literature between 2022 and 2025, in approximate chronological order.

1) Chain-of-Thought (CoT):

Wei et al. [2] introduced Chain-of-Thought prompting, in which the model is conditioned to emit a sequence of intermediate natural-language reasoning steps before producing the final answer. Few-shot CoT exemplars yielded large improvements on arithmetic (GSM8K, SVAMP, MultiArith), commonsense and symbolic-reasoning benchmarks for the PaLM-540B model.

2) Zero-Shot CoT:

Kojima et al. [3] removed the few-shot exemplars entirely, showing that the trigger phrase “Let’s think step by step” suffices to elicit intermediate reasoning in sufficiently large LLMs. This result reframes CoT as an emergent property of the model rather than an artefact of in-context demonstrations.

3) Least-to-Most Prompting:

Least-to-Most prompting decomposes a complex problem into a sequence of progressively simpler sub-problems, solving each in turn and passing its answer to the next stage. On the SCAN compositional-generalisation benchmark this approach reached approximately 99.7 % accuracy when paired with GPT-3, dramatically outperforming standard CoT.

4) Self-Consistency:

Wang et al. [5] proposed Self-Consistency, in which the LLM samples N independent reasoning chains with non-zero temperature and the final answer is determined by majority vote. Reported absolute gains include +10–+18 % on GSM8K, +24 % on MultiArith and +12 % on SVAMP relative to plain CoT. Self-Consistency is the conceptual ancestor of the proposed worklet pipeline.

5) ReAct:

Yao et al. [4] introduced ReAct, which interleaves natural-language “Thought” steps with discrete “Action” calls to external tools (web search, calculators, knowledge-base lookups) and the resulting “Observation” feedback. This thought–action–observation loop materially reduces hallucination on HotpotQA and Fever.

6) Tree-of-Thoughts (ToT):

Yao et al. [6] generalised CoT from a linear chain to a search tree in which the model can branch into multiple candidate next-steps, evaluate each branch with a value function or self-assessment, and back-track when necessary. On the “Game of 24” planning benchmark, GPT-4 solves only about 4 % of instances with plain CoT but approximately 74 % when equipped with ToT, demonstrating the value of explicit search at inference time.

7) Ranked-Voting Self-Consistency:

Wang, Wang and Huang [7] extended Self-Consistency by replacing the plain majority vote with a ranked or weighted vote over candidate answers. Each sampled chain casts a ranked preference, which is then aggregated using a social-choice rule. This variant is more robust on multi-answer and free-form tasks where no single answer attains a clear majority.

8) Combinatorial Reasoning (CR):

Esencan et al. [8] reframed reasoning-fragment selection as a combinatorial-optimisation problem. Their pipeline samples many reasoning chains from an LLM, splits each chain into atomic “reasons,” and constructs a QUBO whose binary variable $x_i \in \{0, 1\}$ decides whether reason i is included in the final prompt. Solving the QUBO yields a task-adaptive prompt without the need for hand-crafted exemplars.

9) Quantum Combinatorial Reasoning (QCR-LLM):

Flores-Garrigós et al. [9] replaced the classical QUBO solver of [8] with a quantum or quantum-inspired annealer and introduced higher-order (HUBO) terms that capture logical coherence between reasons rather than pairwise similarity alone. On reasoning benchmarks the authors report improvements in both accuracy and end-to-end runtime relative to the classical CR baseline. QCR-LLM is the closest published antecedent of the present worklet.

D. Summary

Across the nine algorithms surveyed, two trends are evident. First, the field has moved from generating a single chain (CoT [2], Zero-Shot CoT [3]) toward generating many chains and combining them (Self-Consistency [5], Ranked-Voting SC [7]). Second, combination has progressed from simple voting to explicit combinatorial optimisation over reasoning fragments (CR [8], QCR-LLM [9]). The proposed worklet sits at the end of this trajectory.

IV. QUADRATIC UNCONSTRAINED BINARY OPTIMISATION

A. Definition

A Quadratic Unconstrained Binary Optimisation (QUBO) problem seeks a binary vector $x \in \{0, 1\}^n$ that minimises a quadratic form

$$f(x) = x^T Q x = \sum_i Q_{ii} x_i + \sum_{i < j} Q_{ij} x_i x_j \quad (1)$$

where Q is a symmetric (or, equivalently, upper-triangular) cost matrix. The diagonal entries Q_{ii} represent linear costs on each variable, while the off-diagonal entries Q_{ij} represent pairwise interaction costs. QUBO is NP-hard in general because it generalises Max-Cut, but a wide variety of practical combinatorial problems—scheduling, routing, portfolio selection, feature subset selection, graph colouring and, most importantly for this paper, reasoning-fragment selection—map naturally onto it.

B. Mapping Reasoning Selection onto QUBO

Following Esencan et al. [8], one variable x_i is introduced for each candidate reasoning fragment produced by the SLM. Two effects are encoded:

- 1) *Relevance*: for each fragment i , we set $Q_{ii} = -\text{relevance}(i, \text{query})$, so that switching x_i on lowers the objective when the fragment is relevant.
- 2) *Redundancy*: for each pair (i, j) of fragments, we set $Q_{ij} = +\text{redundancy}(i, j)$, so that selecting two near-duplicate fragments increases the objective.

Minimising $f(x)$ therefore selects a subset of reasoning fragments that is simultaneously relevant and non-redundant—precisely the desideratum for a compact, high-quality reasoning prompt.

C. Solving QUBO

Three families of solvers are relevant for this worklet. For small instances ($n \lesssim 20$), exhaustive enumeration over the 2^n binary vectors is tractable. For larger instances, classical heuristics such as Simulated Annealing yield good approximate solutions in polynomial expected time. Finally, quantum and quantum-inspired annealers, including D-Wave hardware and the open-source dwave-neal Simulated Annealing Sampler, accept the QUBO as their native input language and have been the subject of intensive recent study [9], [10]. Chandarana et al. [10] demonstrate digital quantum-optimisation algorithms that achieve a measurable runtime advantage over the best classical heuristics on certain QUBOs, motivating a future migration of the worklet pipeline from the simulated to the genuinely quantum regime.

V. POSITIONING THE PROPOSED WORKLET PIPELINE

The pipeline targeted by Samsung PRISM worklet 26ARS05 consists of six stages: (1) sample N reasoning chains from an SLM; (2) split each chain into atomic reasoning fragments; (3) build a QUBO from query-relevance and pairwise redundancy; (4) solve the QUBO using a quantum-inspired Simulated Annealing sampler; (5) inject the selected fragments back into the SLM as a refined CoT prompt; and (6) produce the final answer via a second SLM forward pass.

Each stage has a clear antecedent in the literature reviewed above. Stage 1 corresponds to the sampling step of Self-

Consistency [5]; stages 2–3 mirror the fragment-level QUBO construction of Combinatorial Reasoning [8]; stage 4 generalises the classical CR solver in the direction of Quantum Combinatorial Reasoning [9]; and stages 5–6 amount to a standard CoT prompt followed by a second decoding pass [2].

The novelty of the proposed worklet, relative to this prior art, lies in three points. First, the entire pipeline is targeted at SLMs in the 3 B parameter regime rather than at frontier LLMs, with the explicit goal of recovering reasoning quality lost to compression. Second, the QUBO solver is realised by dwave-neal on commodity CPUs and on Nvidia A100 HPC nodes, providing a deployment-ready intermediate point between purely classical CR and the genuinely quantum QCR-LLM. Third, the implementation exposes a one-line switch between the simulated and the quantum back-end, so the same pipeline can be re-run on D-Wave hardware once the necessary access is provisioned.

VI. CONCLUSION

This paper has surveyed three closely interrelated bodies of work: large-language-model reasoning, recent variants of Chain-of-Thought prompting and Quadratic Unconstrained Binary Optimisation. The review motivates an architecture in which an SLM produces many candidate reasoning fragments, a QUBO encodes the relevance–redundancy trade-off among those fragments, and a quantum-inspired annealer selects an optimal subset for a refined CoT prompt. Subsequent worklet activities will instantiate this architecture on Llama-3.2-3B with HPC support and benchmark its accuracy and runtime against Self-Consistency [5] and standard CoT [2] baselines.

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